WITH HIGH PROBABILITY THERE IS NO SMALL SEPARATOR WHOSE COMPLEMENT HAS BETWEEN U_1 AND N/2 VERTICES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned author source sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The thresholds u_0 and u_1 of the section and their relation, together with the expansion estimate for sets in the product graph. Set up in the author's section on Tutte obstructions and the size thresholds (source lines 655-700); the expansion estimate is the author's displayed inequality eq:expansion-log. Supplies the two size bounds that the proof converts into exponential estimates, and the expansion used in the small-boundary case.

Source reference. Original equation 1.1: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 224–224. Navigation pointer only: the counted proof uses this displayed expansion estimate in its small-boundary case; the display belongs to an earlier section and is not reproduced in this card.

Source reference. Original lemma 6.3: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 714–714. Navigation pointer only: the counted proof uses the author's counting lemma for sets with a small boundary; that lemma has its own card in this lane.

Lemma 0.1. *Whp, there are no disjoint sets $U, B \subseteq V$ satisfying*

$$|U| \le u_0, \qquad u_1 \le |B| \le n/2,$$

for which every edge from B to $V \setminus (B \cup U)$ is not present in G_{p_-}.

Proof. Denote by $b \coloneqq |B|$. The number of possible U is at most

$$\sum_{u \le u_0} \binom{n}{u} \le \exp\left\{u_0 \ln(en/u_0)\right\} = \exp\left\{u_0 \cdot \frac{M^5}{p_-} \cdot \ln d\right\} \le \exp\left\{\frac{b}{\ln d}\right\},$$

where we used the definition of u_0 and u_1 and that $b \ge u_1$.

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If $e_G(B, B^c) \geq b \ln^2 d$, then at least $b \ln^2 d - d|U| \geq b \ln^2 d/2$ edges from B avoid U. Thus, the probability that there are sets U, B with $b = |B|$ as in the claim is at most

$$\binom{n}{b} \exp\left\{\frac{b}{\ln d}\right\} (1 - p_-)^{b \ln^2 d/2} = o(n^{-3}).$$

If $e_G(B, B^c) < b \ln^2 d$, Original lemma 6.3 gives at most $\exp\{2b/\ln d\}$ choices for B. Moreover the expansion property in (1.1) gives $e_G(B, B^c) \geq \alpha_M b$ for some constant α_M while $du_0 = o(b)$, thus at least $\frac{\alpha_M}{2} b$ edges from B avoid U. Thus, the probability that there are sets U, B with $b = |B|$ as in the claim is at most

$$\exp\left\{\frac{3b}{\ln d}\right\} (1 - p_-)^{\alpha_M b/2} \leq \exp\{-\alpha'_M b\}.$$

A union bound over $b \geq u_1$ concludes the proof. □